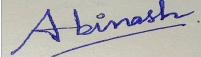


TPDS	TEST REPORT																																																								
BRAKE THERMAL MAPPING HSD, GHD, CDT CONDITION TEST ON P201 VP PHASE VEHICLE IN GVW CONDITION																																																									
To : NAGRIKAR RAVI/MISHRA KRISHNA		Report No: TPDS/VT/RDP201/VP/2024/27987																																																							
Project No: RDP201-UPP Goods-G2-Domestic		Report Date: 30-May-2024																																																							
Objective: To evaluate brake thermal mapping test on P201 VP phase vehicle.																																																									
Scope: To conduct brake thermal mapping test on P201 VP phase vehicle in GVW condition																																																									
Vehicle Configuration:																																																									
Foundation Brakes	BIL	Booster Size	9"																																																						
Front Brake	C60, Single pot	Booster Ratio	5																																																						
Front Disc Size, Eff radius	277 x 24, 116	Jump in	12 +/-3																																																						
Front Brake Pad Grade	Apache 5409M	Pedal Ratio	3.5:1																																																						
Rear Brake	25.4 mm (WC)	TMC Size	23.81																																																						
Rear Drum size, Eff. radius	279 X 50.8, 279	Stroke-Primary/Secondary	17.5/18.5mm																																																						
Rear Brake Pad Grade	RD0014	System Split	"X" Split																																																						
Test: 06-C-06-1339-001-BRAKE THERMAL MAPPING CITY CONDITION																																																									
Observation: Refer the table below for test observations:-																																																									
<table><tr><th colspan="2">P201 BRAKE TEMPERATURE MAPPING</th><th>CDT</th><th>GHD</th><th>HSD</th></tr><tr><td colspan="2">Distance Covered (km)</td><td>149</td><td>243</td><td>264</td></tr><tr><td colspan="2">No. of Shift operated</td><td>3</td><td>3</td><td>3</td></tr><tr><td colspan="2">No. of Brake application per km</td><td>6.7</td><td>3.2</td><td>1.8</td></tr><tr><td rowspan="2">Front pad peak temperature (deg C)</td><td>Average</td><td>104</td><td>104</td><td>82</td></tr><tr><td>Maximum</td><td>163</td><td>179</td><td>188</td></tr><tr><td rowspan="2">Rear Liner peak temperature (deg C)</td><td>Average</td><td>94</td><td>98</td><td>72</td></tr><tr><td>Maximum</td><td>153</td><td>153</td><td>146</td></tr><tr><td colspan="2">Average speed (kmph)</td><td>26</td><td>40</td><td>49</td></tr><tr><td colspan="2">Maximum speed (kmph)</td><td>70</td><td>80</td><td>80</td></tr><tr><td colspan="2">Brake Energy Killed (J/kg-km)</td><td>62.3</td><td>61.9</td><td>38.2</td></tr></table>					P201 BRAKE TEMPERATURE MAPPING		CDT	GHD	HSD	Distance Covered (km)		149	243	264	No. of Shift operated		3	3	3	No. of Brake application per km		6.7	3.2	1.8	Front pad peak temperature (deg C)	Average	104	104	82	Maximum	163	179	188	Rear Liner peak temperature (deg C)	Average	94	98	72	Maximum	153	153	146	Average speed (kmph)		26	40	49	Maximum speed (kmph)		70	80	80	Brake Energy Killed (J/kg-km)		62.3	61.9	38.2
P201 BRAKE TEMPERATURE MAPPING		CDT	GHD	HSD																																																					
Distance Covered (km)		149	243	264																																																					
No. of Shift operated		3	3	3																																																					
No. of Brake application per km		6.7	3.2	1.8																																																					
Front pad peak temperature (deg C)	Average	104	104	82																																																					
	Maximum	163	179	188																																																					
Rear Liner peak temperature (deg C)	Average	94	98	72																																																					
	Maximum	153	153	146																																																					
Average speed (kmph)		26	40	49																																																					
Maximum speed (kmph)		70	80	80																																																					
Brake Energy Killed (J/kg-km)		62.3	61.9	38.2																																																					
Conclusion: Brake thermal mapping HSD,GHD,CDT test on P201 VP phase vehicle data generated.																																																									
Recommendation: Nil																																																									
Attachments: Data, Graph																																																									
Test Conducted / Reported by		Reviewed by	Approved by																																																						
																																																									
Abinash Abinash		MITTAL SAPAN	Rathakrishnan M																																																						
CC SHABBIR KARNURE/LALASURE SANTOSH																																																									

TPDS		VEHICLE CONFIGURATION			
Project No : RDP201-UPP Goods-G2-Domestic				Report No : TPDS/VT/RDP201/VP/2024/27987	
Vehicle Details					
Parameter		P201 VP UP78 DMT Vehicle			
Model		G2 3D15 2550 195R15 40L FSD			
Reg No / S No		R6B15023			
Engine Model & No		1.5 lit, 3 Cyl CUR6B44576			
C.G.Height(MM)		902 mm			
Transmission /Gearbox		5 speed			
Drive		2WD			
Axle Ratio		--			
Wheel / Tyre Details		Tubeless 195 R15 UL: 44 psi , 44 psi, L:44 psi, 65 psi CEAT			
Front Suspension		Rigid-Semi Elliptical (5 Leaf)			
Rear Suspension		Rigid-Semi Elliptical (2 stage -8			
Steering Details		RCBT			
Speedo Make/Raio		--			
Odometer					
Wheel Track(mm)		Front - 1465, Rear - 1375			
Wheel Base(mm)		2550			
F/R Ratio		--			
Dyn.Rolling Radius		--			
Weight Transfer Ratio		--			
Weight Distribution:					
UNLADEN	Curb Wt (kg) FAW (kg) RAW (kg) ETW (kg)	1631 1157 474 1631			
LADEN	GVW (kg) FAW (kg) RAW (kg) Pay Load (kg)	3038 1400 1638 1260			
OVERLOAD	GVW (kg) FAW (kg) RAW (kg) Pay Load (kg)	1407 1495 2305 3800			
Amb Temperature		34 Deg C			
Wind Velocity		<3m/s			
Gust		< 5 m/s			
Rel. Humidity		--			
Weather		Sunny			
Road Surface		Tar			
Instruments Used		Link logger & L350			
Test Site		MSPT			
Test Driver		Prabha			
Test Engineer		Abinash			
Test Date		03-04-2024 - 09-04-2024			

UPP_G2_P201- BRAKE DUTY CYCLE MAPPING ROUTE – CDT GHD HSD

MSPT Validation

10th May 2024

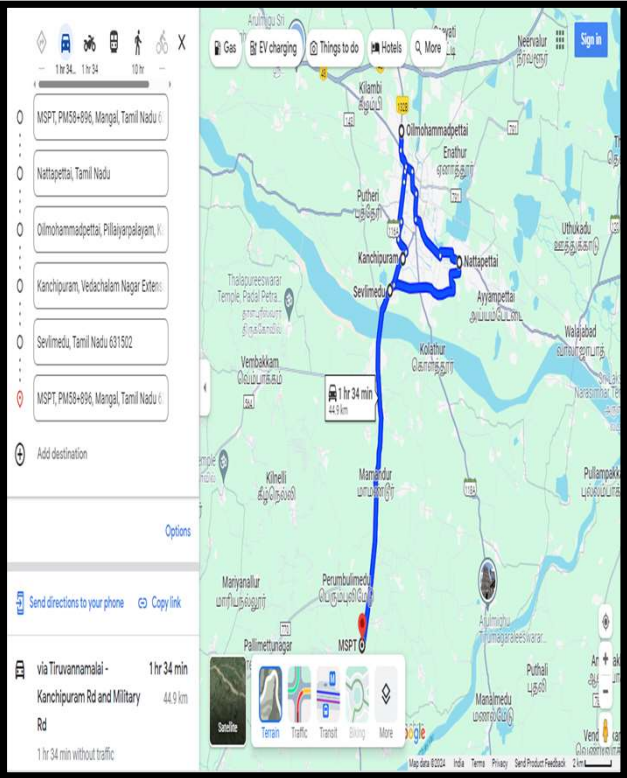
VEHICLE & TEST DETAILS

PARAMETERS	P201_UP78
Engine	1.5 lit, 3 Cylinder, BS 6.2
Max Power	80hp
Max Torque	210Nm
Transmission	5MT320 (Bolero Pass./TUV300) with change for UPP from Direct shift to 5 speed cable shift.
Tyre Make & Size	CEAT 195 R15 Tubeless
Foundation Brakes System	Brakes India Limited
Brake split type	X
Software ID	NA
Load Condition	FAW: 1400 , RAW : 1638 , GVW : 3038
Vehicle Operation	Back to back
Test Drivers	Sridhar, Sivanantham

BRAKE SYSTEM CONFIGURATIONS

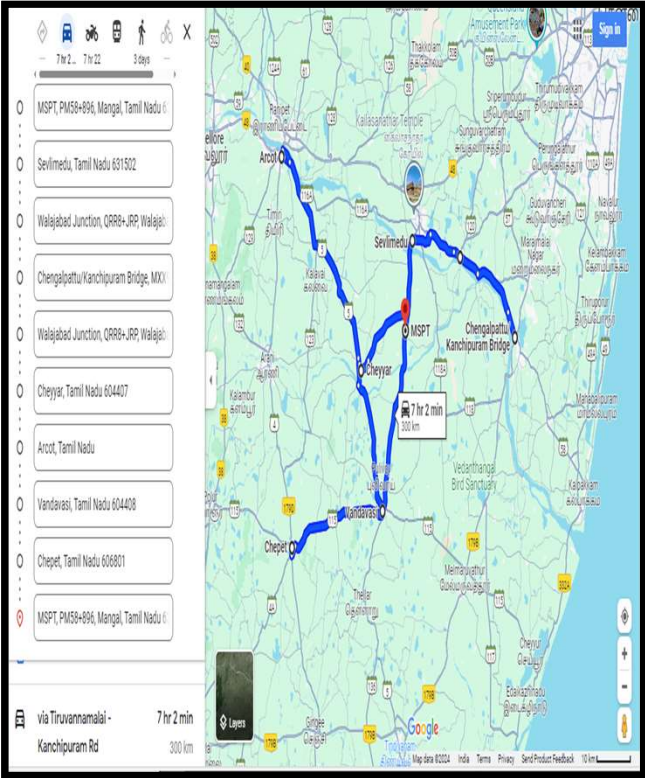
Description		UoM	P201_UP78
Front Brake	Caliper Size	mm	60
	Eff. Rad	mm	116
	Pad Grade	-	Apache 5409M
	Pad Mue	-	0.42
	Disc Size	mm	24
Rear Brake	Drum dia (size)	mm	279 X 50.8 (Bolero carryover)
	Eff. Rad	mm	279
	Liner Grade	-	RD0014i4
	Wheel Cylinder Dia	mm	25.4
BTMC	Booster Size	Inch	9
	Booster Ratio	-	5
	Jump-In	bar	12 +- 3 bar
	TMC Size	mm	23.81
	Stroke (Pri/Sec)	mm	17.5/18.5
Pedal	Pedal Ratio	-	3:5:1

ROUTE DETAILS



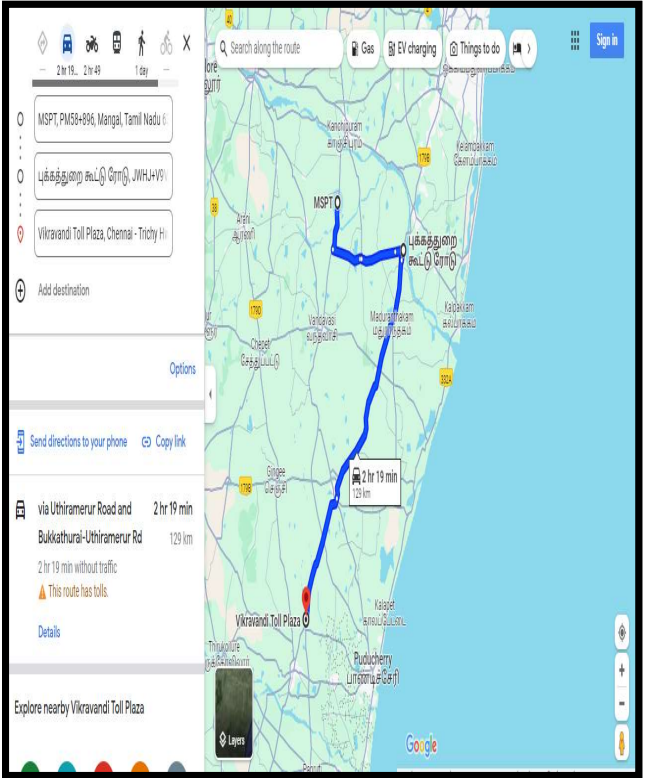
CDT

MSPT - Sevilimedu - Periyar Nagar- Taluk office
- Kanchipuram Bus stand - Olimuhamad pettai
- Keelambi bypass - Return MSPT



GHD

MSPT – Sevilimedu – Walajabad – Chengalpet –
Sevilimedu – Mangal – Cheyyar – Arcot -
Cheyyar-Vandavasi – Chetpet - Return MSPT



HSD

MSPT – Uthiramerur – Melmaruvathur –
Tindivanam – Vikravandi toll - Return MSPT

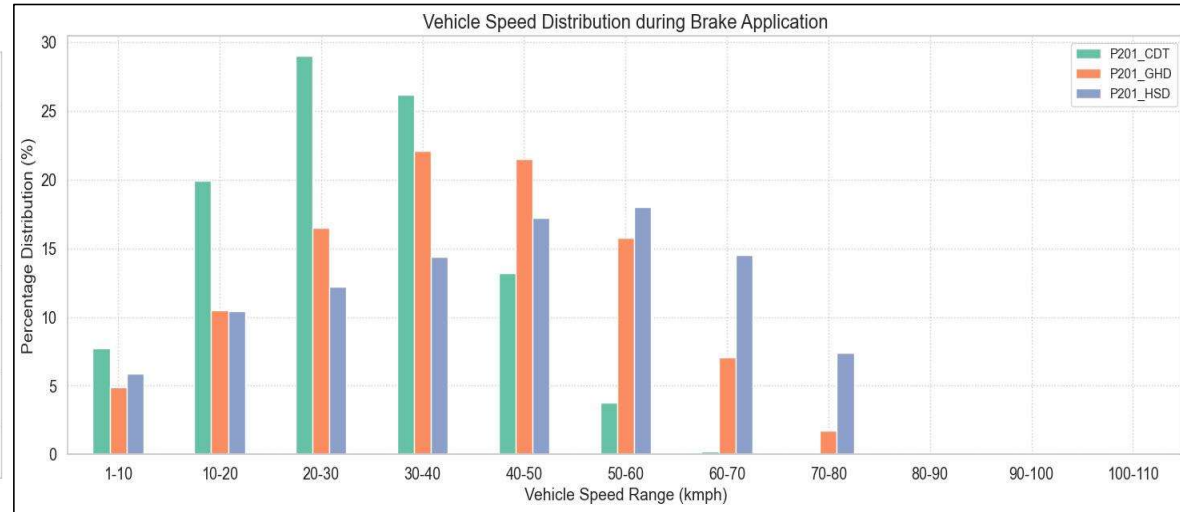
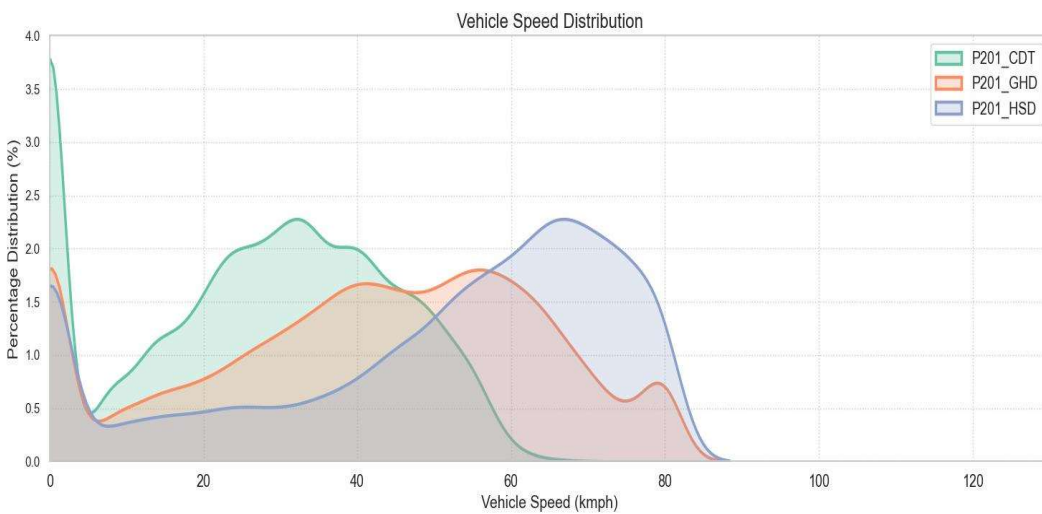
SUMMARY

P201 BRAKE TEMPERATURE MAPPING		CDT	GHD	HSD
Distance Covered (km)		149	243	264
No. of Shift operated		3	3	3
No. of Brake application per km		6.7	3.2	1.8
Front pad peak temperature (deg C)	Average	104	104	82
	Maximum	163	179	188
Rear Liner peak temperature (deg C)	Average	94	98	72
	Maximum	153	153	146
Average speed (kmph)		26	40	49
Maximum speed (kmph)		70	80	80
Brake Energy Killed (J/kg-km)		62.3	61.9	38.2

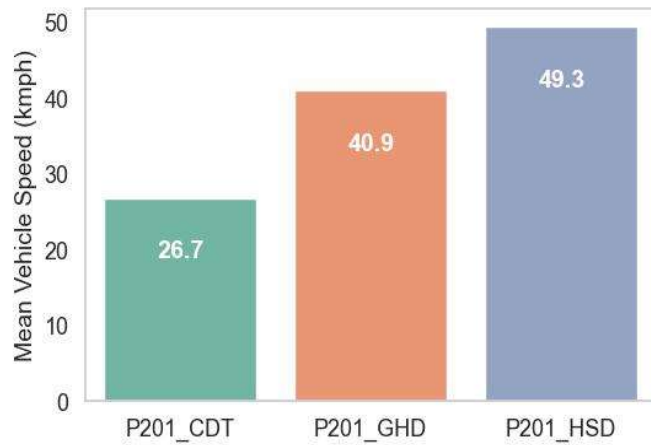
SUMMARY

P201 BRAKE TEMPERATURE MAPPING		CDT	GHD	HSD
Distance Covered (km)		149	243	264
No. of Shift operated		3	3	3
No. of Brake application per km		6.7	3.2	1.8
Front pad peak temperature (deg C)	Average	104	104	82
	Maximum	163	179	188
Rear Liner peak temperature (deg C)	Average	94	98	72
	Maximum	153	153	146
Average speed (kmph)		26	40	49
Maximum speed (kmph)		70	80	80
Brake Energy Killed (J/kg-km)		62.3	61.9	38.2

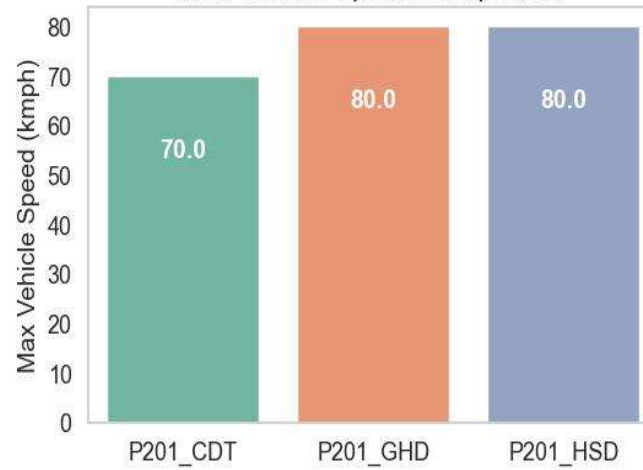
VEHICLE SPEED COMPARISON



Mean Vehicle Speed Comparison



Max Vehicle Speed Comparison



StdDev Vehicle Speed Comparison

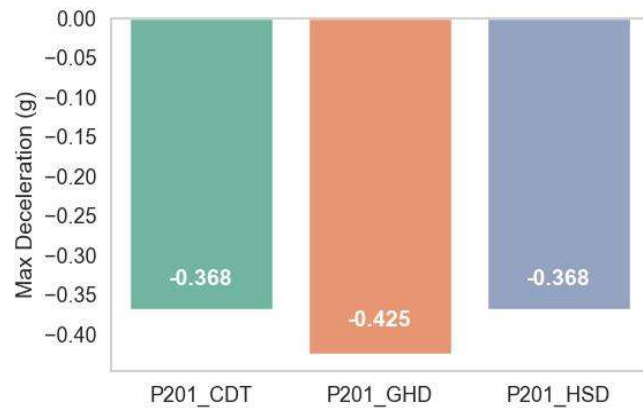


VEHICLE DRIVING PATTERN COMPARISON

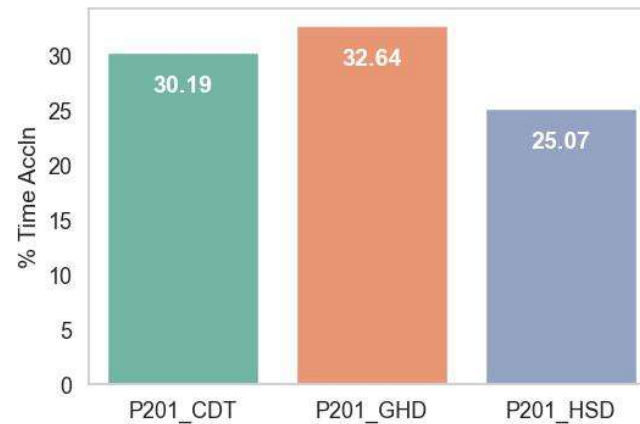
Mean Positive Longitudinal Accln Comparison



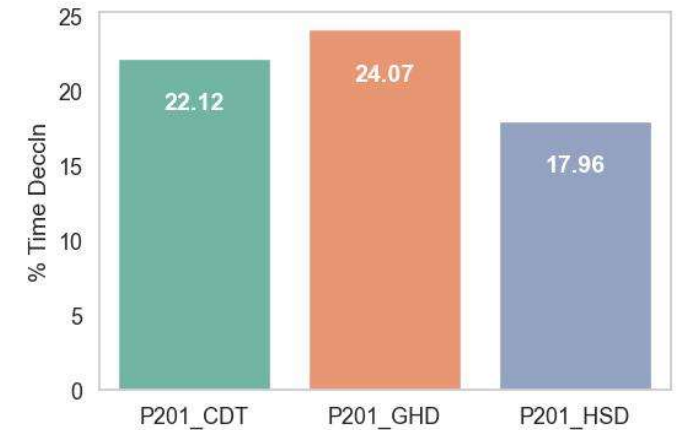
Max Longitudinal Deceleration Comparison



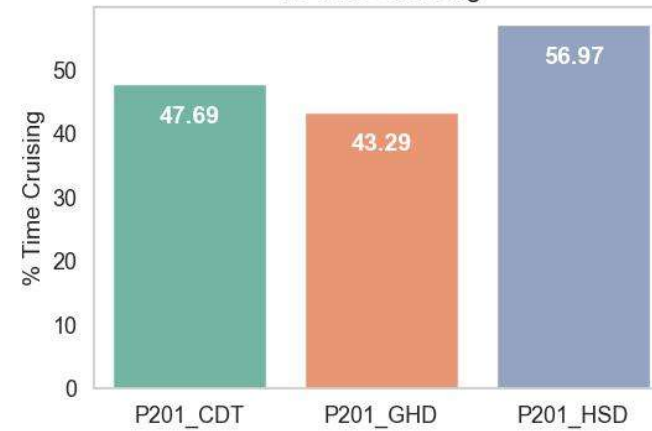
% Time Accln



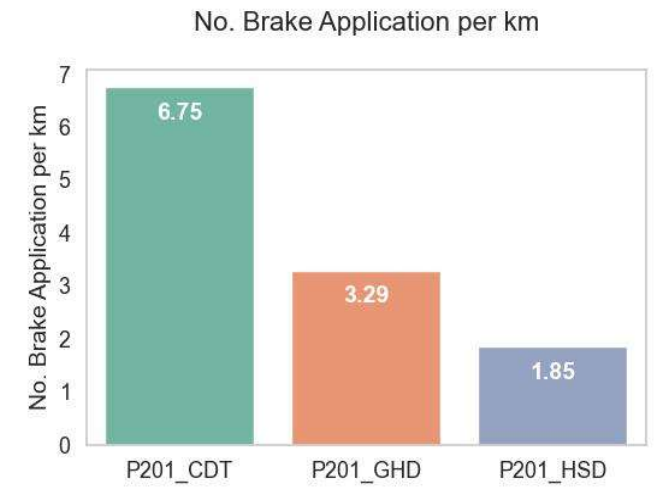
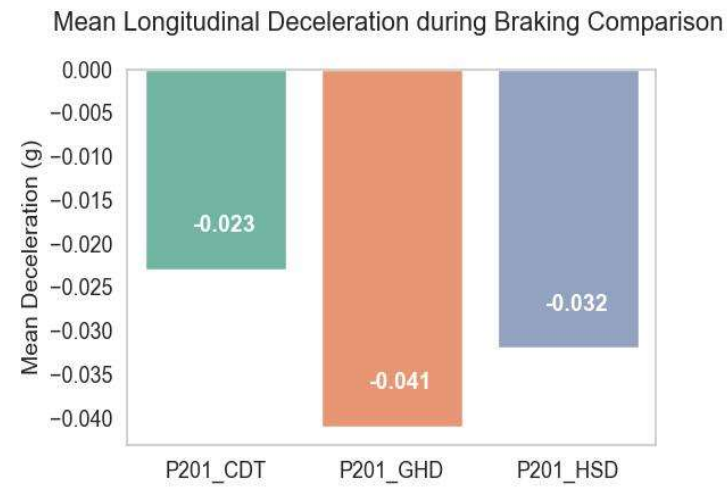
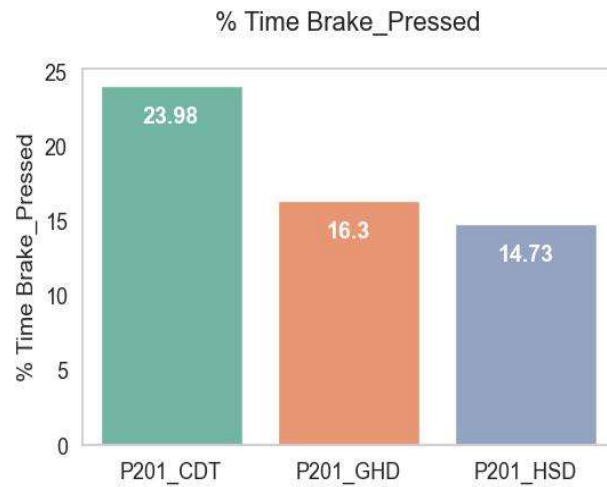
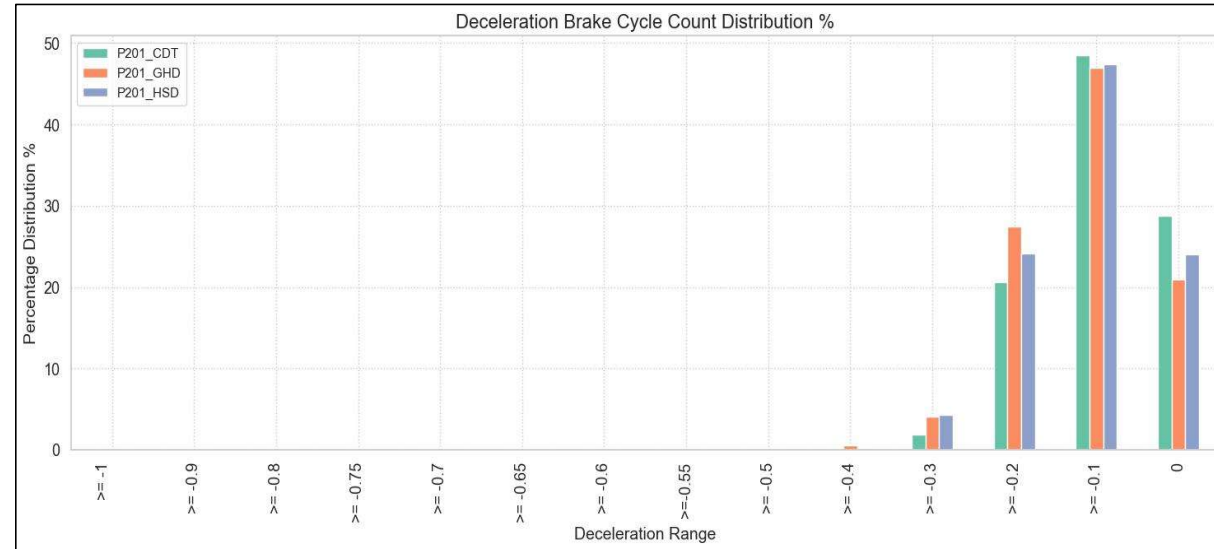
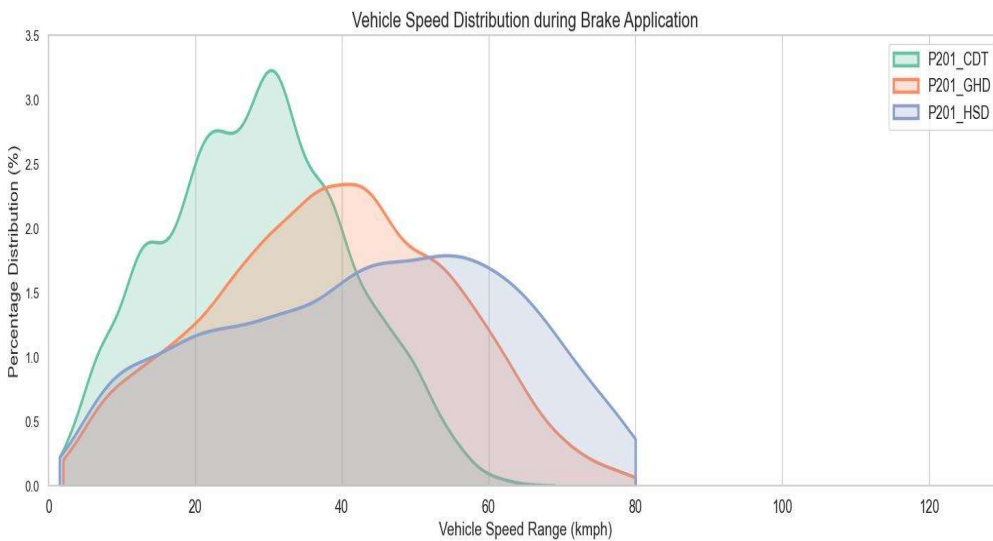
% Time Deccln



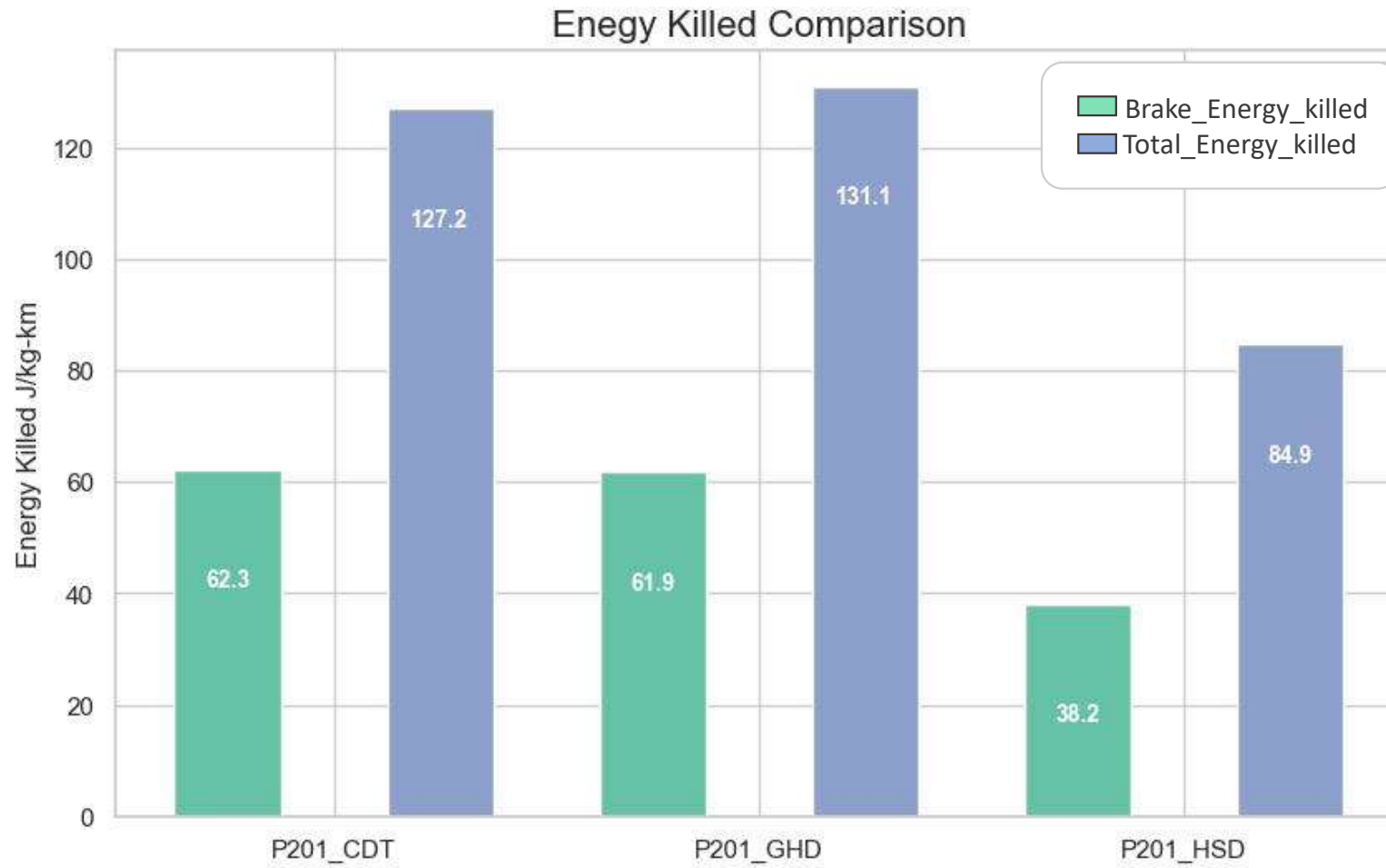
% Time Cruising



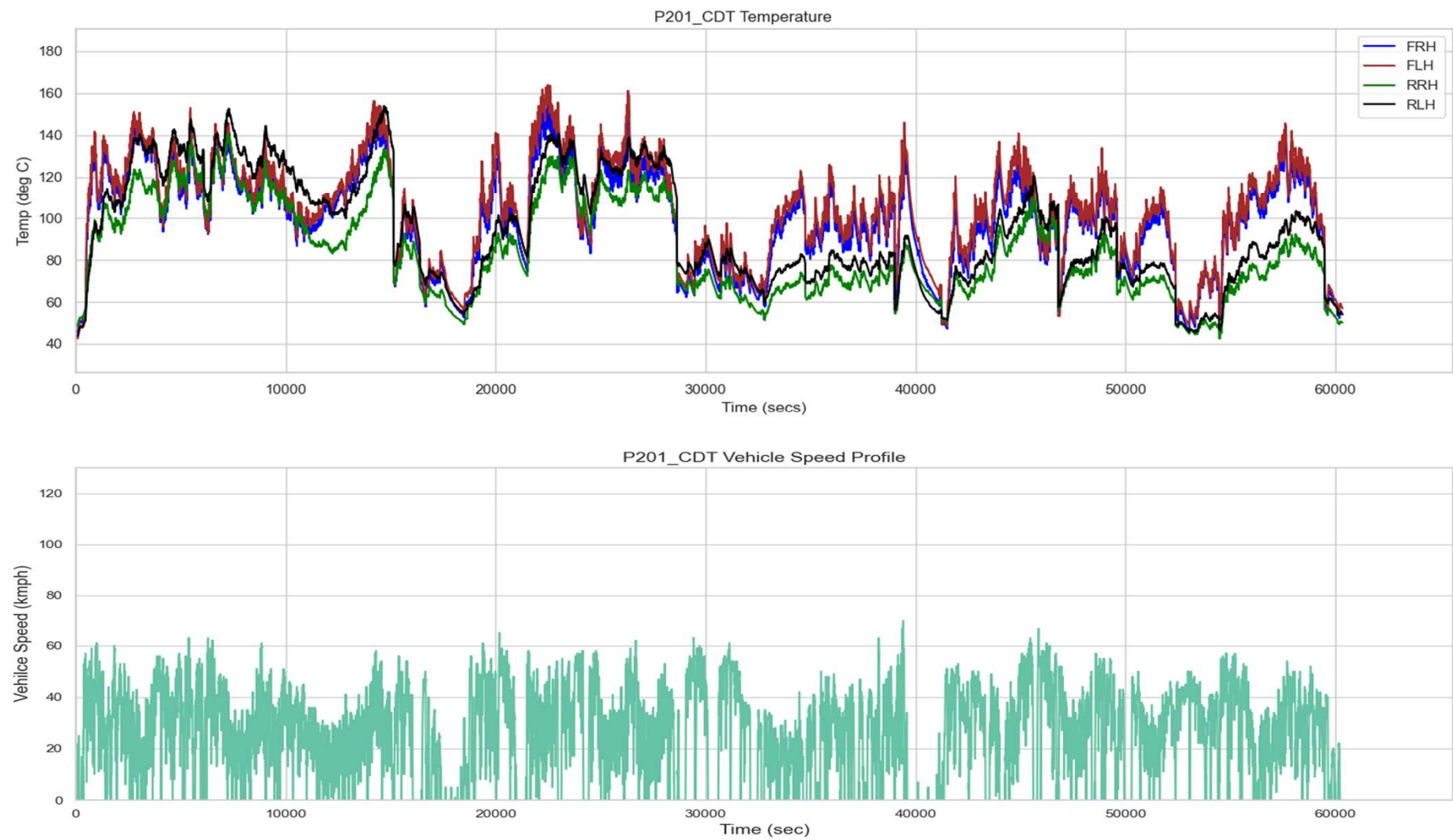
BRAKE PROFILE COMPARISON



ENERGY KILLED COMPARISON



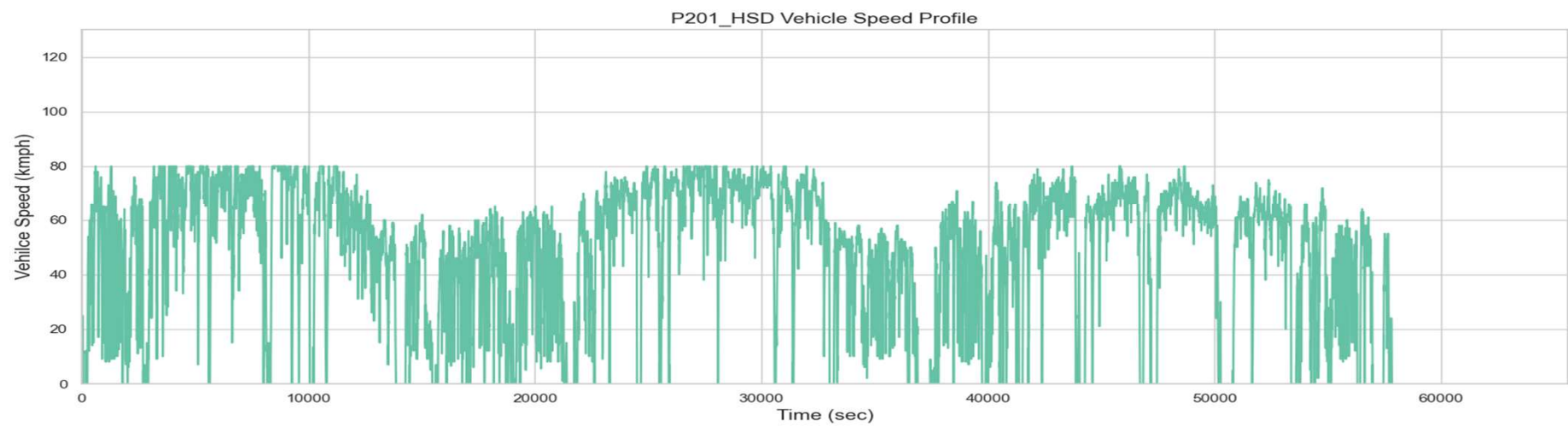
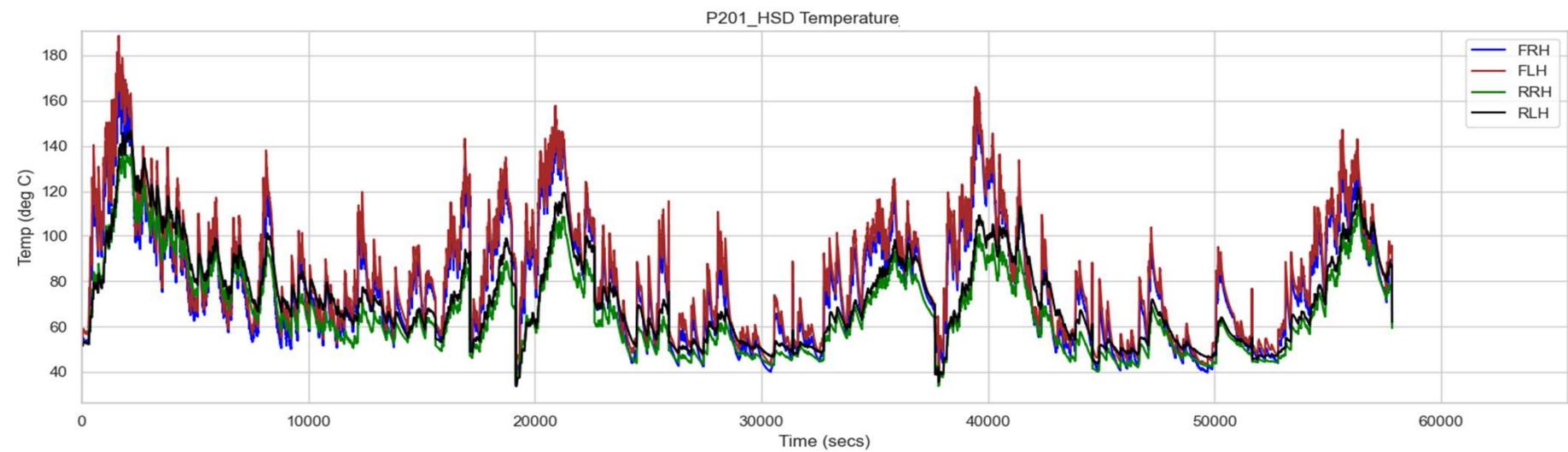
BRAKE TEMPERATURE PROFILE - CDT



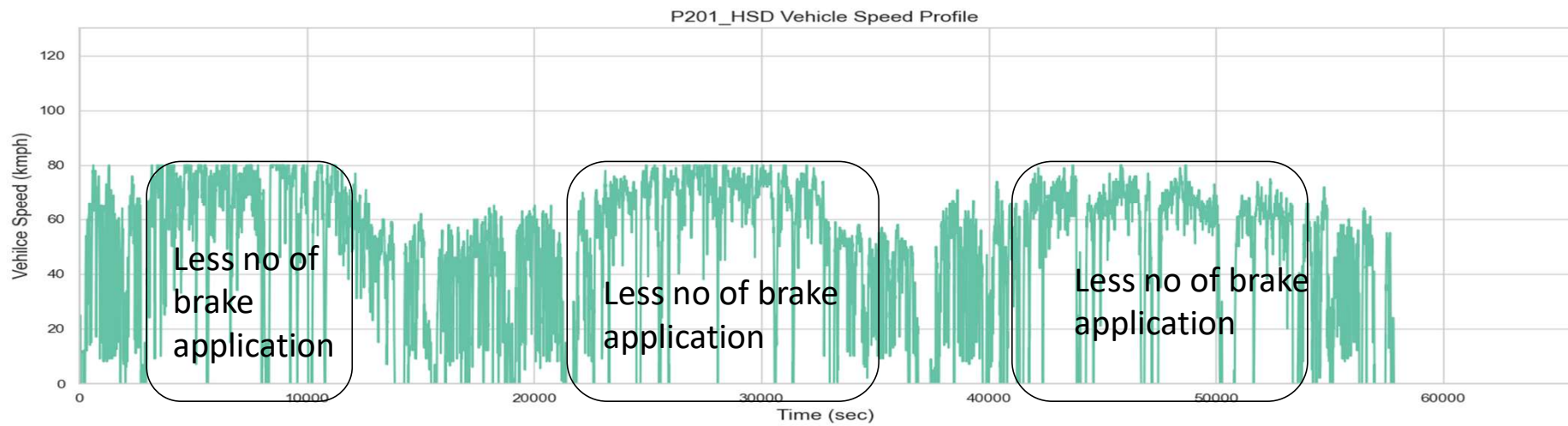
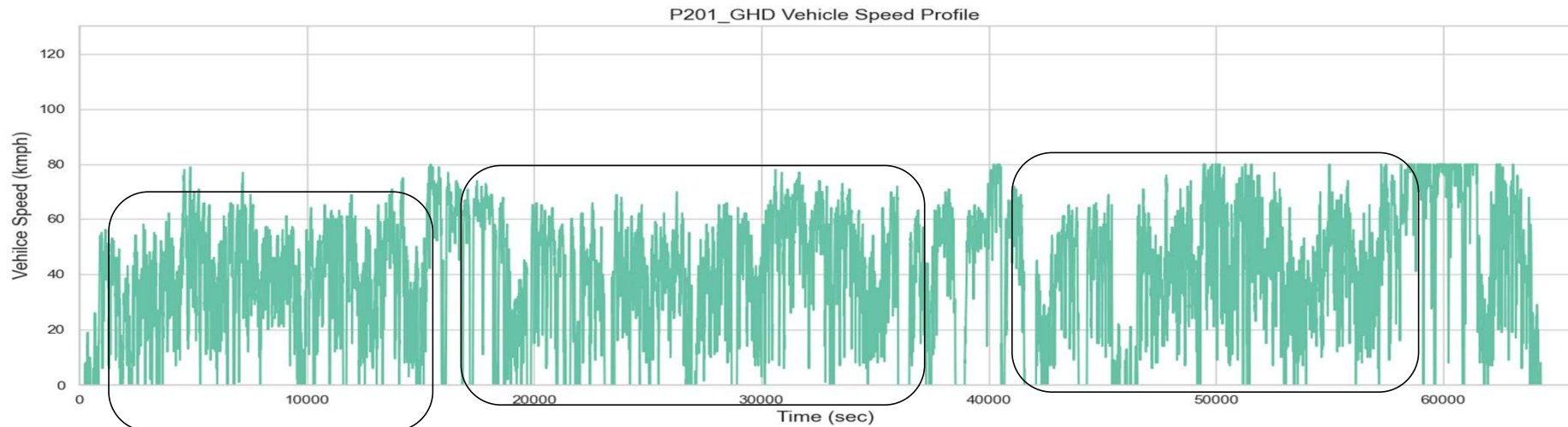
BRAKE TEMPERATURE PROFILE - GHD



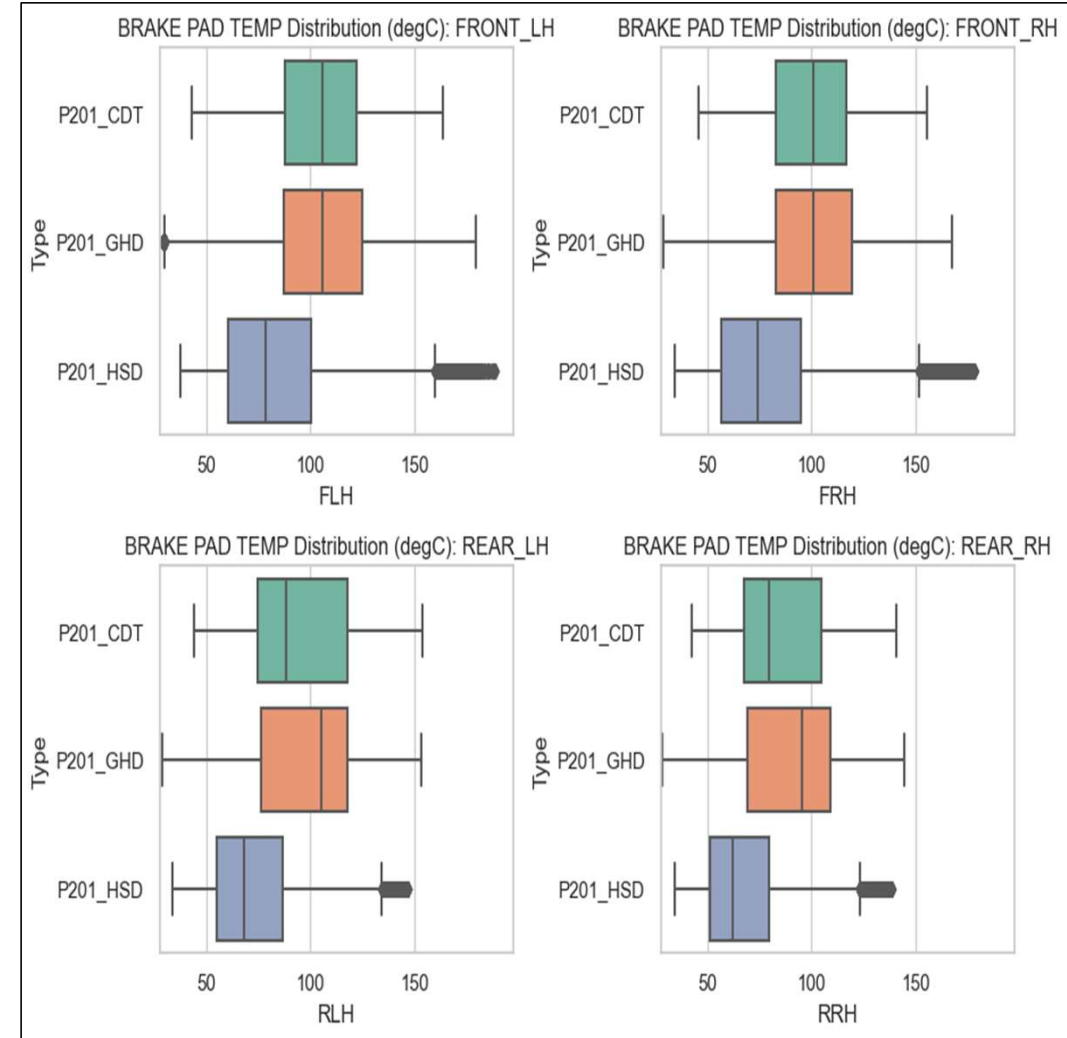
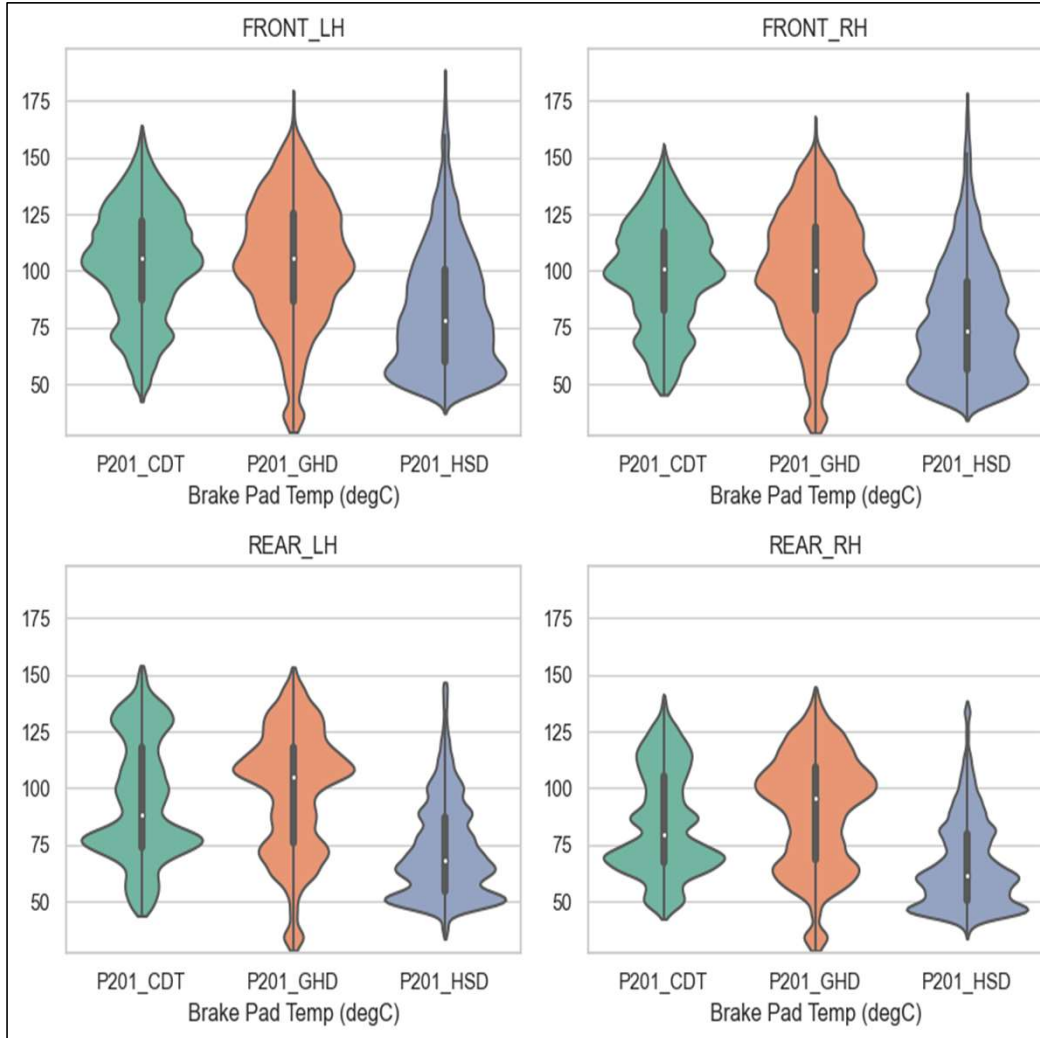
BRAKE TEMPERATURE PROFILE - HSD



BRAKE TEMPERATURE PROFILE - HSD



BRAKE TEMPERATURE DISTRIBUTION



CONCLUSION

- ❖ Energy killed during brake application are 62.3 J/kg-km , 61.9 J/kg-km & 38.2 J/kg-km for CDT, GHD &HSD respectively.
- ❖ No. of brake application is 6.7 brake/km, 3.2 brake/km & 1.8 brake/km for CDT, GHD & HSD respectively.
- ❖ Front maximum brake temperature are 163, 179 & 188°C for CDT, GHD & HSD respectively.
- ❖ Rear maximum brake temperature are 153, 153 & 146 °C for CDT, GHD & HSD respectively.